

Laplace Transform

Script: Chapter 7 (Laplace transform, transfer functions, stability)

Learning goals

- Describe engine sensors by transfer functions $H(s)$; relate poles and zeros to decay time, resonance frequency, damping, Q and bandwidth.
- Compute impulse and step responses by partial fractions and decide stability from the pole locations.
- Read and sketch Bode diagrams; judge an RC anti-aliasing filter.
- Handle a transport delay e^{-sT_d} by Padé approximation and understand the consequences of a right-half-plane zero.
- Verify analytic results by numerical ODE integration (Euler, RK4) and understand the stability limit of explicit integrators.
- Map an analog system into the z -domain (forward Euler, impulse invariance, bilinear transform) – a preview of Sheets 9 and 10.

Background

For a causal signal $x(t)$ the (unilateral) Laplace transform is $X(s) = \int_0^\infty x(t) e^{-st} dt$. An LTI system described by a linear ODE with constant coefficients has the *transfer function*

$$H(s) = \frac{Y(s)}{X(s)} = \frac{b_0 s^m + \dots + b_m}{a_0 s^n + \dots + a_n} = K \frac{\prod_k (s - z_k)}{\prod_k (s - p_k)},$$

the Laplace transform of its impulse response $h(t)$. A causal system is BIBO stable iff all poles satisfy $\text{Re } p_k < 0$. On the imaginary axis, $s = j\omega$, $H(j\omega)$ is the frequency response.

The knock burst of the virtual engine (`code/common/engine_signals.h`) is

$$x_k(t) = A e^{-t/\tau} \sin(2\pi f_d t) \varepsilon(t), \quad f_d = 6.5 \text{ kHz}, \quad \tau = 0.8 \text{ ms}$$

(plus a second mode at 10.5 kHz). We interpret it as the impulse response of the mechanical path *combustion chamber* $\rightarrow$ *engine block* $\rightarrow$ *piezo sensor*, which is excited by the short pressure shock of knocking combustion. Throughout this sheet we use the normalised second-order model ($\sigma = 1/\tau$, $\omega_d = 2\pi f_d$)

$$H_K(s) = \frac{\omega_0^2}{s^2 + 2\sigma s + \omega_0^2}, \quad \omega_0^2 = \omega_d^2 + \sigma^2, \quad \zeta = \sigma/\omega_0. \quad (1)$$

Exercise 7.1 Sensor models in the s -domain

PAPER

- Knock resonator.** Show that the impulse response of (1) is $h_K(t) = \frac{\omega_0^2}{\omega_d} e^{-\sigma t} \sin(\omega_d t) \varepsilon(t)$. Compute σ , ω_0 (and f_0), ζ , the quality factor $Q = 1/(2\zeta)$, the poles and the region of convergence. Is the system stable? What would $\sigma < 0$ mean?
- Compute the **step response** of H_K with partial fractions. Give the overshoot, the time of the first maximum and the 2% settling time. The DC gain is $H_K(0) = 1$: what does this imply for the slow combustion vibration (a half sine over 60° crank angle) that is also present on the sensor signal?
- Bode diagram.** Sketch $|H_K(j\omega)|$ in dB and the phase over $\log f$ (asymptotes!). Compute the height and frequency of the resonance peak and show that the -3 dB bandwidth is $\Delta\omega \approx 2\sigma$, i.e. $B \approx 1/(\pi\tau)$ in Hz. Interpret this relation between decay time and bandwidth.

- d) The second knock mode has $f_d = 10.5$ kHz and the same τ . Mark all four poles of the two-mode model in the s -plane. What distinguishes the poles of the two modes, and what do they have in common?
- e) **RC anti-aliasing filter.** The knock signal is sampled at $f_s = 40$ kHz behind an RC low-pass with $R = 1$ k Ω , $C = 10$ nF. Give $H_{RC}(s)$, its pole and f_c . Compute the attenuation at 6.5 kHz, 10.5 kHz and at 30 kHz. Onto which frequency does a 30 kHz component alias, and why is that critical? Which f_c would a first-order filter need for 40 dB attenuation at 30 kHz? Compare with a second-order Butterworth filter with the same f_c and conclude. Also give the phase and the group delay of the RC filter at 6.5 kHz.
- f) **Lambda sensor with transport delay.** Fuel change $\rightarrow$ sensor is modelled as $G(s) = e^{-sT_d}/(1 + sT)$ with $T_d = 200$ ms (gas travel) and $T = 30$ ms (sensor), the values of the Wokwi lambda-probe chip.
- Why is $G(s)$ not a rational function, and why is this a problem for the methods of this chapter?
 - Derive the Padé(1,1) approximation $e^{-sT_d} \approx (1 - sT_d/2)/(1 + sT_d/2)$ by matching the Taylor series of both sides as far as possible. Where are poles and zeros of the approximated $G_1(s)$?
 - Compute the step response of $G_1(s)$ with partial fractions. Determine the minimum (time and value) and explain the *undershoot*. Compare with the exact step response.
 - Phase error: the exact delay has phase $-\omega T_d$, the Padé term $-2 \arctan(\omega T_d/2)$. Up to which frequency is the error below 5° ? (Use the series $\arctan x \approx x - x^3/3$.)

Exercise 7.2 Knock resonator: ODE solvers vs. Laplace

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Program `code/jslinux/sheet07/resonator_ode.c` integrates the ODE $\ddot{y} + 2\sigma\dot{y} + \omega_0^2 y = \omega_0^2 u$ of (1) in state-space form $x = (y, \dot{y})^T$.

- Implement `rhs()`, one step of the forward Euler method and one step of the classical Runge–Kutta method (RK4).
- Implement the analytic impulse and step responses of Exercise 7.1. The program simulates the impulse response (initial state $y(0) = 0$, $\dot{y}(0) = \omega_0^2$ – why?) and the step response for step sizes $h = 40, 20, \dots, 0.625$ μ s and prints the maximum error. Estimate the order of convergence p from successive errors ($e(h) \propto h^p$).
- Explain the results of the Euler method: derive the stability condition $|1 + h\lambda| < 1$ for $\dot{x} = \lambda x$ and evaluate it for the poles of H_K . Compare the limit with the sampling period of the engine chip (20 μ s). Why does the chip compute the burst from the closed-form solution instead of integrating an ODE?
- Implement `freq_resp()` and compare peak, resonance frequency and -3 dB bandwidth with your results of Exercise 7.1 c).

Exercise 7.3 Lambda sensor: delay and Padé approximation

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Program `code/jslinux/sheet07/lambda_pade.c` compares the exact step response of $G(s) = e^{-sT_d}/(1 + sT)$ with three rational approximations of the delay: a first-order lag $1/(1 + sT_d)$, Padé(1,1) and Padé(2,2)

$$e^{-sT_d} \approx \frac{1 - sT_d/2 + s^2T_d^2/12}{1 + sT_d/2 + s^2T_d^2/12}.$$

- Any strictly proper $G(s) = N(s)/D(s)$ can be simulated in *controllable canonical form*: $D(p)v = u$ and $y = N(p)v$, where $p = d/dt$ and state $(v, \dot{v}, \dots)$. Implement `ccf_rhs()` and `ccf_out()` (integration by RK4 is given).

- b) Implement the exact system by a delay line (ring buffer of $D = T_d/h$ samples) followed by the exact discretisation of the lag, $y_{k+1} = a y_k + (1 - a) u_{k-D}$, $a = e^{-h/T}$, and the analytic step response. Enter the Padé coefficients. Compare minimum, time of $y = 0.5$, RMS and maximum error of the three approximations.
- c) Compare the phase of the delay part. Up to which frequency is the phase error of Padé(1,1) and Padé(2,2) below 5° ?
- d) Which approximation would you use (i) to design a lambda controller, (ii) to predict the sensor signal in the time domain?

Exercise 7.4 From s to z : discretisation of analog systems

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A digital filter shall imitate $H_K(s)$ and $H_{RC}(s)$ at $f_s = 40$ kHz, $T = 1/f_s$. Three classical mappings are

$$\begin{aligned} \text{forward Euler: } s &= \frac{z-1}{T}, & \text{bilinear: } s &= \frac{2}{T} \frac{z-1}{z+1}, \\ \text{impulse invariance: } h[n] &= T h(nT) \quad (\text{poles } z_k = e^{p_k T}). \end{aligned}$$

- a) (**paper**) Into which region of the z -plane is the left half s -plane mapped by each method, and onto which curve the imaginary axis? Which method can turn a stable analog system into an unstable digital one? Show that the bilinear transform maps ω_a onto $\Omega = 2 \arctan(\omega_a T/2)$ (*frequency warping*).
- b) Complete the three mapping functions of the program `code/jslinux/sheet07/discretise.c`, i.e. `map_euler`, `map_bilin` and `impinv` (second-order section $H(s) = (b_0 s^2 + b_1 s + b_2)/(a_0 s^2 + a_1 s + a_2)$; impulse invariance via partial fractions with the small complex struct provided).
- c) *Prewarping*: with $s = K \frac{z-1}{z+1}$, $K = \omega_r / \tan(\omega_r T/2)$, the frequency ω_r is mapped exactly. Use $\omega_r = \omega_0$ for the resonator and $\omega_r = \omega_c$ for the RC filter.
- d) Run the program and interpret the tables: pole radius and angle, peak frequency, gain at DC and close to $f_s/2$. Which method would you choose for the knock resonator, which for the anti-aliasing filter model?

Deliverables

Paper solution of 7.1 including the Bode sketch; the error table and the order estimate of 7.2 with the Euler stability limit; the step-response table of 7.3 and your recommendation 7.3 d); the comparison tables of 7.4 with a short justification of your choice of discretisation method.